

Christophe Hatterer

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SUMMARY

MSc candidate in Computational Science & Engineering (ETH Zürich) with an MIT thesis on mixed-variable Bayesian optimisation for engineering design. Builds structured, tested scientific Python — packaged libraries with pytest, ruff, and mypy — and contributed 3D generative models to EngiOpt, an open-source engineering-design optimisation library, through a maintainer-reviewed pull request. Proficient in Python (NumPy, SciPy, pandas, PyTorch) and C/C++ (with OpenMP), with experience running large-scale experiments on SLURM-managed HPC clusters. Comfortable translating mathematical and statistical models into clean, maintainable software.

EDUCATION

Massachusetts Institute of Technology - MIT - DeCoDe Lab <i>MSc Thesis in Mechanical Engineering - Supervised by Prof. Faez Ahmed and Prof. Mark Fuge</i> Benchmarking Transferability of Geometric Intelligence in Generative Engineering Models	Mar. 2026 – Exp. Sep. 2026 Boston, MA, USA
(Swiss) Federal Institute of Technology in Zurich - ETH Zürich <i>MSc in Mechanical Engineering, specialisation Computational Science and Engineering</i> Relevant coursework: Probabilistic Artificial Intelligence; Optimization and Machine Learning; Towards Efficient and High-Performance Computing for Engineers (code versioning, testing, OOP, parallelisation); Engineering Design Optimization.	Sep. 2024 – Exp. Sep. 2026 Zürich, Switzerland
(Swiss) Federal Institute of Technology of Lausanne - EPF Lausanne <i>BSc in Mechanical Engineering</i> Specializing in Applied Mathematics	Sep. 2020 – Sep. 2023 Lausanne, Switzerland
Technical University of Delft - TU Delft <i>Student Exchange Year</i> Specializing in Aerospace and Aeronautical Engineering	Aug. 2022 – Jun. 2023 Delft, Netherlands
Island School <i>International Baccalaureate</i> Specialization Maths, Physics, Chemistry, French	Jun. 2019 Hong Kong, Hong Kong SAR

PROFESSIONAL EXPERIENCE

Multi-Physics Simulations Engineer <i>Hitachi Energy Ltd. - Internship</i> - Built multi-physics COMSOL models of self-healing behaviour in HV transformer capacitors, with Python post-processing and analysis tooling to turn simulation output into design recommendations. - Lead-authored an internal research paper documenting the multi-physics modelling framework.	Aug. 2025 – Feb. 2026 Baden-Daetwill, Switzerland
Aerospace Engineer <i>Axalp Technologies - Internship</i> - Developed dynamic simulation models for aircraft landing-gear kinematics in transient load cases. - Led end-to-end development (scoping to integration testing) of a lightweight thermal cooling system for an aircraft auxiliary battery pack in Siemens NX, achieving ~30% weight reduction.	Sep. 2023 – Jun. 2024 Olten, Switzerland
Project Assistant Director <i>Asia One Technics Ltd. - Internship</i> Coordinated procurement, supplier quality control, and on-site inspections for a solar charging-canopy installation.	Jun. 2019 – Sep. 2019 Hong Kong, Hong Kong

RESEARCH

Research Assistant <i>Computational Mechanics Group, ETH Zürich</i> - Built a Python data-processing pipeline on SLURM-managed HPC clusters, using open-source research modules to analyse large X-ray computed-tomography datasets of mortar specimens. - Studied fracture propagation in heterogeneous materials via X-ray CT with in-situ testing and digital volume correlation. - Designed components for a torsional crack-propagation experiment. - Supervised by Prof. Laura De Lorenzis (ETH) and Dr. Pietro Carrara (ETH).	Sept. 2024 – Aug. 2025 Zürich, Switzerland
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- Extended generative models (GANs, SliceGAN, and a multi-variable Variational Autoencoder) to the 3D domain for engineering-design generation.
- Contributed the 3D generative models to **EngiOpt**, an open-source engineering-design optimisation library, through a maintainer-reviewed pull request (IDEAL-Lab/EngiOpt #43, ~2,100 lines); conformed code to the project's **ruff**, **mypy**, and pre-commit checks.
- Supervised by Prof. Mark Fuge and Dr. Florian Felten.

SKILLS & EXPERTISE

Programming	Scientific Python	Software & Tooling	Languages
Python, C/C++	NumPy, SciPy, pandas	Git, pytest, ruff, mypy	French (Native)
Julia, Matlab	PyTorch, BoTorch / GPyTorch	pip / hatch packaging, Docker	English (Native)
Fortran, VBA	Bayesian Optimisation	Hydra, Weights & Biases	German (B1)
PostgreSQL, OpenMP	Machine Learning / Generative AI	SLURM / HPC clusters	Mandarin (Limited)

PROJECTS

- **Bayesian Optimisation of drug-candidate features** — optimised predicted bioavailability while maintaining synthesizability constraints.
- **High-performance C++ N-body simulation** (Barnes-Hut) — reduced computational complexity from $O(n^2)$ to $O(n \log n)$ and parallelised with **OpenMP**.
- **Bayesian Neural Network with SWA-Gaussian** for satellite image classification — quantified predictive uncertainty to reduce false positives.
- **Gaussian Process model** for environmental PM2.5 prediction in residential areas.
- **Python transient analysis of landing-gear structural dynamics** — added numerical-dissipation schemes, improving simulation stability by 45%.

Volunteering

- Nepal Service Trip – Orphanage Support – Kathmandu, Nepal (2018).
- HK Asia Pacific Adventure – Sports Instructor – Hong Kong (2017).
- Vietnam Service Trip – Library Construction – Hanoi, Vietnam (2016).

LEADERSHIP ROLES & EXPERIENCES

Mechanical System Advisor

Jul. 2024 – Jan. 2025
Zürich, Switzerland

aCentauri Solar Racing

- Advising 20+ team members across Aerodynamics, Mechanical, and Structures for BWSC 2025, a 3000km solar endurance race.
- Implementing GANTT-based project scheduling, budgeting, and systems engineering methodologies.
- Enhancing solar car Simcenter GFEM for precise load case simulations and optimized vehicle development.

Lead Chassis Engineer

Oct. 2021 – Jun. 2022
Maassluis, Netherlands

NOVA Electric Racing

- Led a team of 6 in prototyping an electric racing bike chassis for performance.
- Developed sustainable flax-fiber composite bodywork using DFM techniques.
- Conducted FEM analysis in Ansys Workbench for 20+ components, validating the motorcycle frame's structural integrity.

Finance Manager

Oct. 2021 – Jan. 2023
Lausanne, Switzerland

PolySI - Integration Week

- Managed budgets and logistics for events, ensuring smooth execution for incoming students.